

Non-Stationary Time Series

EC 421, Set 9

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Prologue

Schedule

Last Time

Autocorrelation

Today

A brief introduction to nonstationarity

Upcoming

Assignment!

Nonstationarity

Nonstationarity

Intro

Let's go back to our assumption of **weak dependence/persistence**

Nonstationarity

Stationarity

Requirements for **stationarity** (a *stationary* time-series process):

Nonstationarity

Random walks

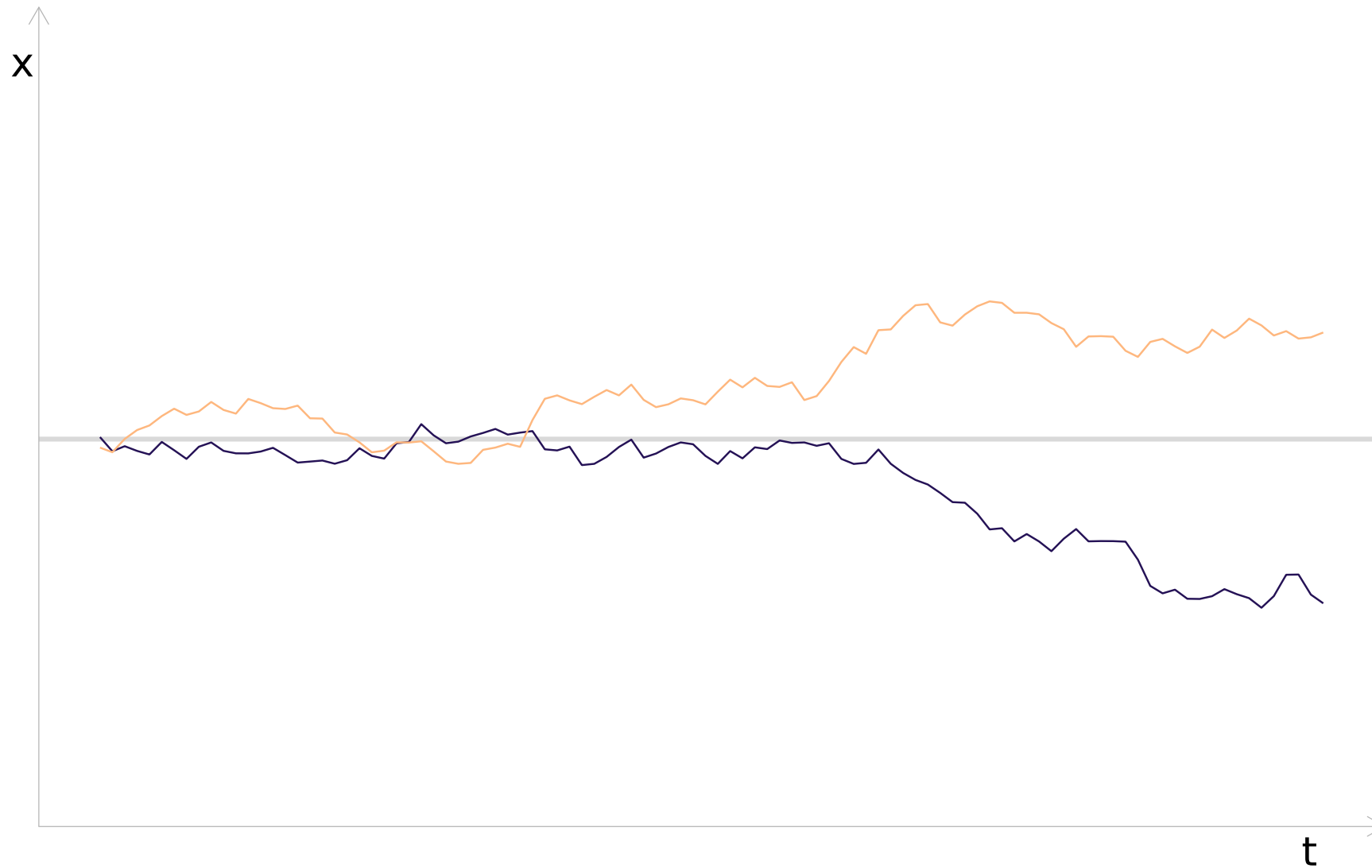
Random walks are a famous example of a nonstationary process:

What's the big deal?

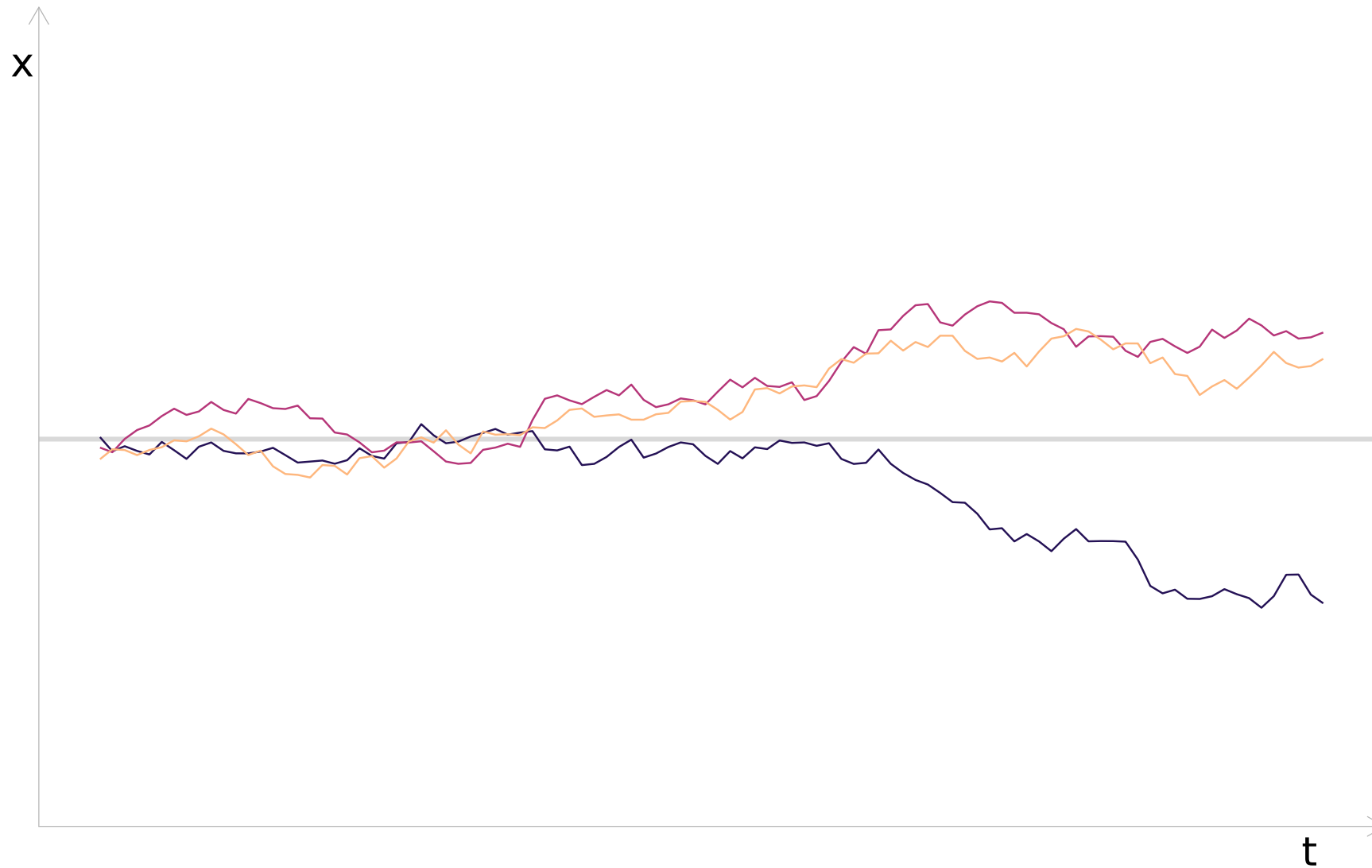
One 100-period random walk



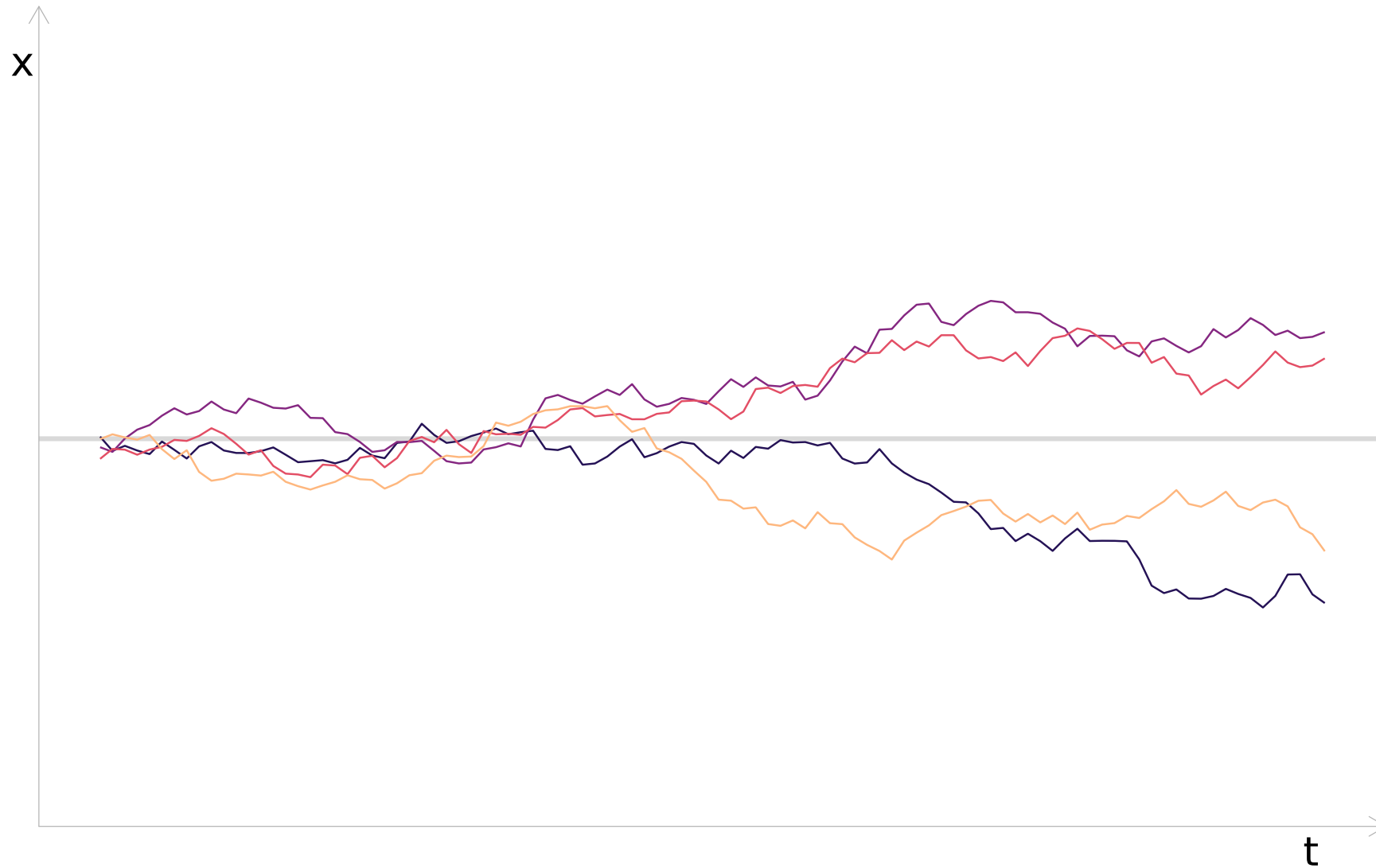
Two 100-period random walks



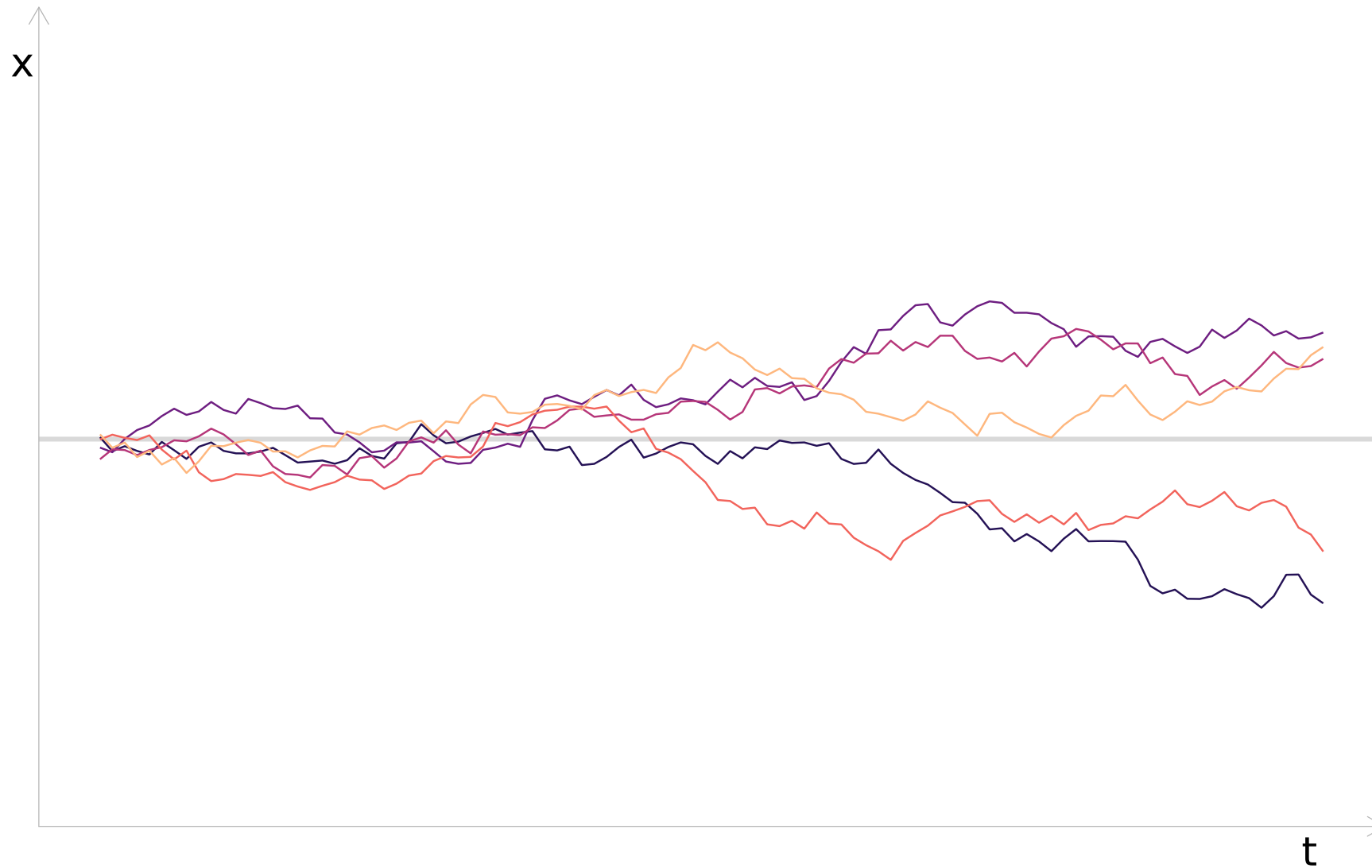
Three 100-period random walks



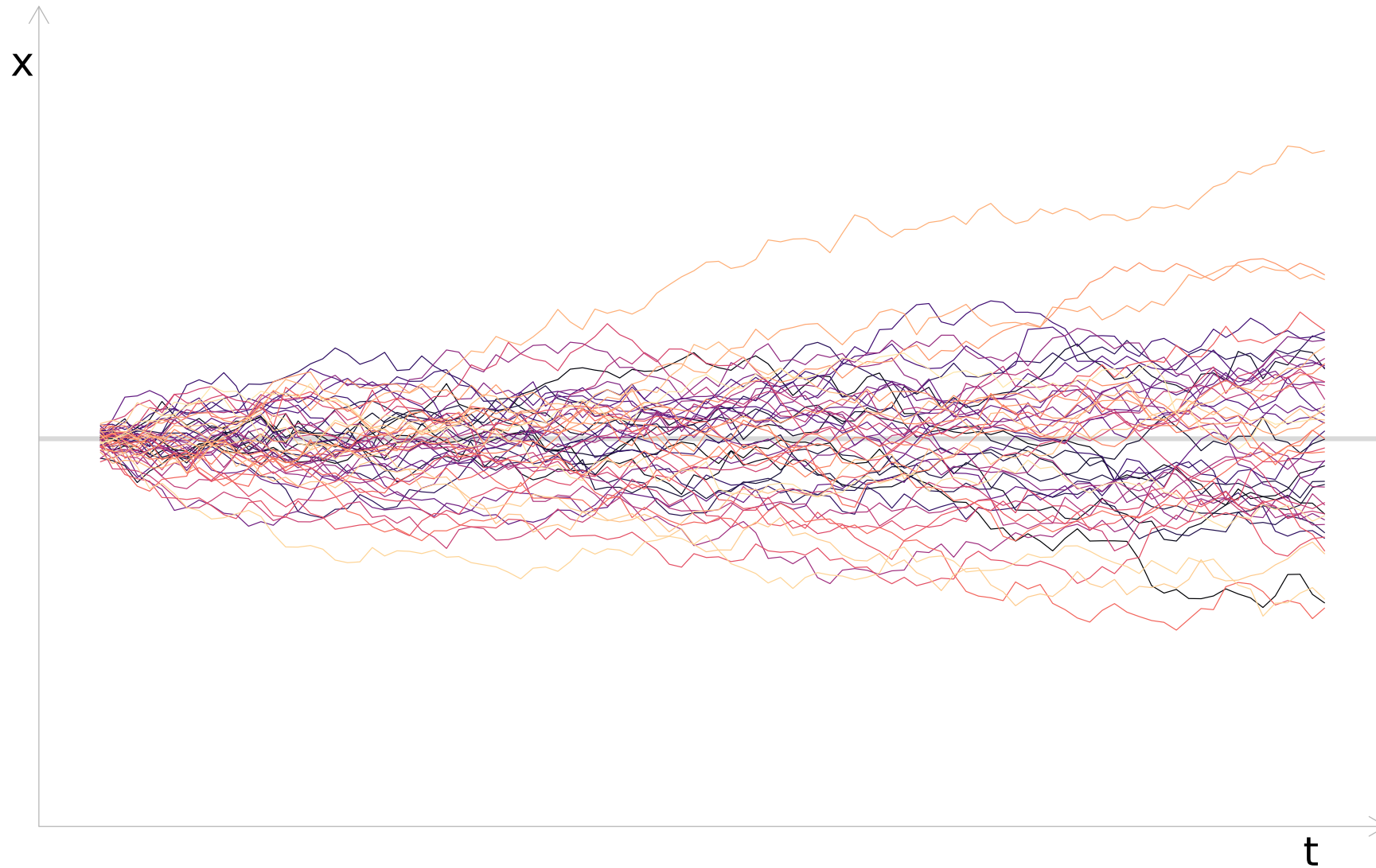
Four 100-period random walks



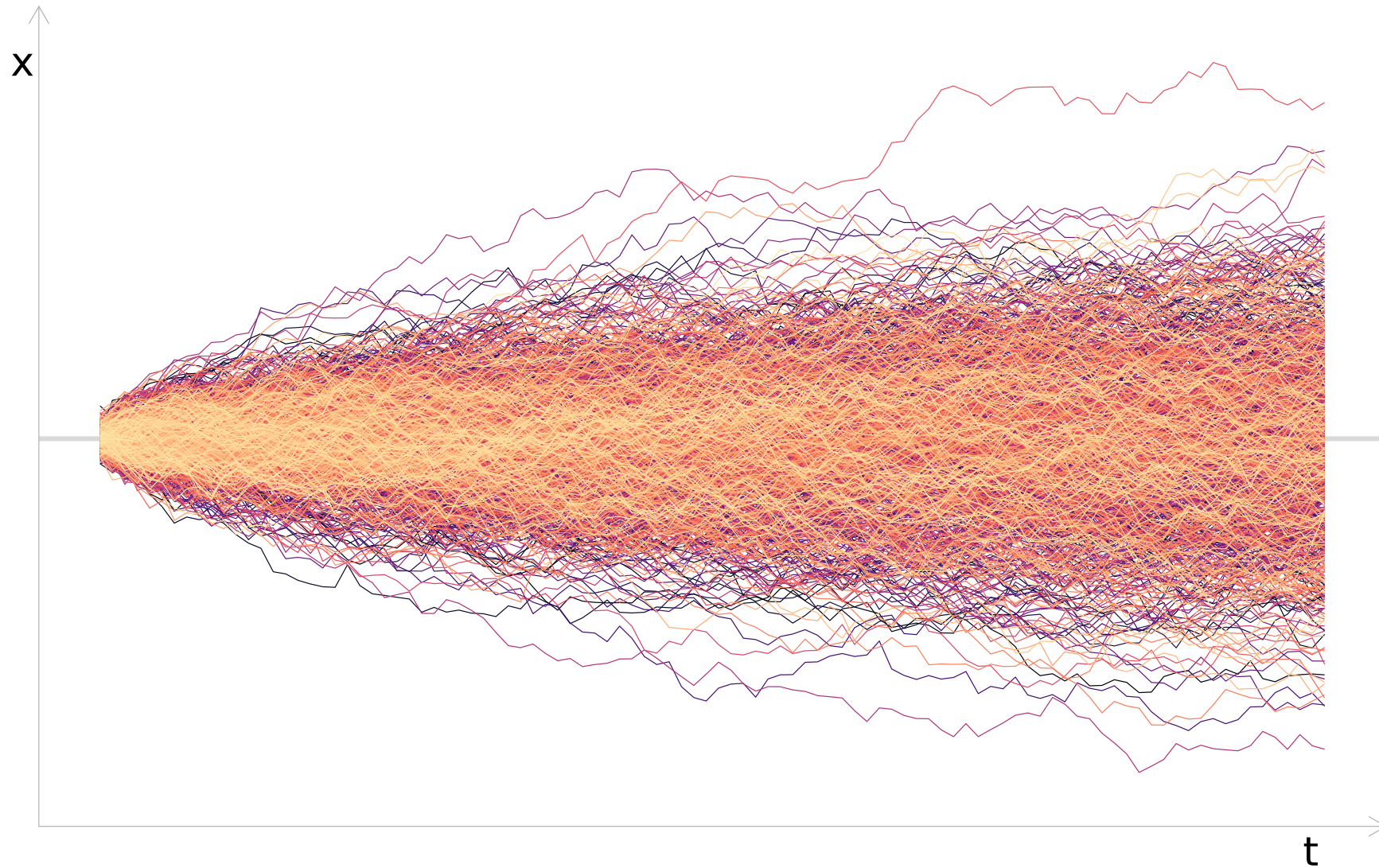
Five 100-period random walks



Fifty 100-period random walks



1,000 100-period random walks



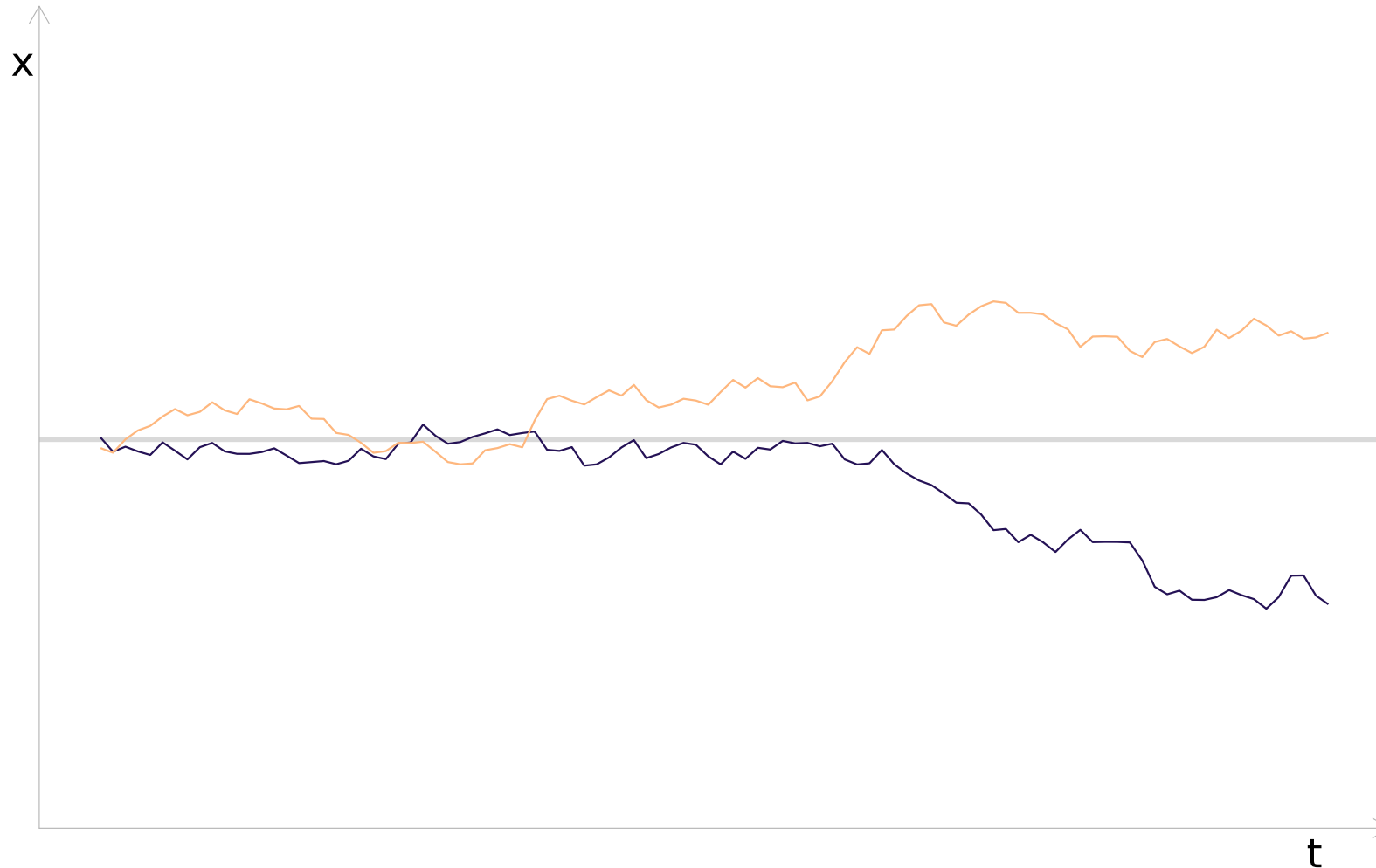
Nonstationarity

Problem

One problem is that nonstationary processes can lead to **spurious** results.

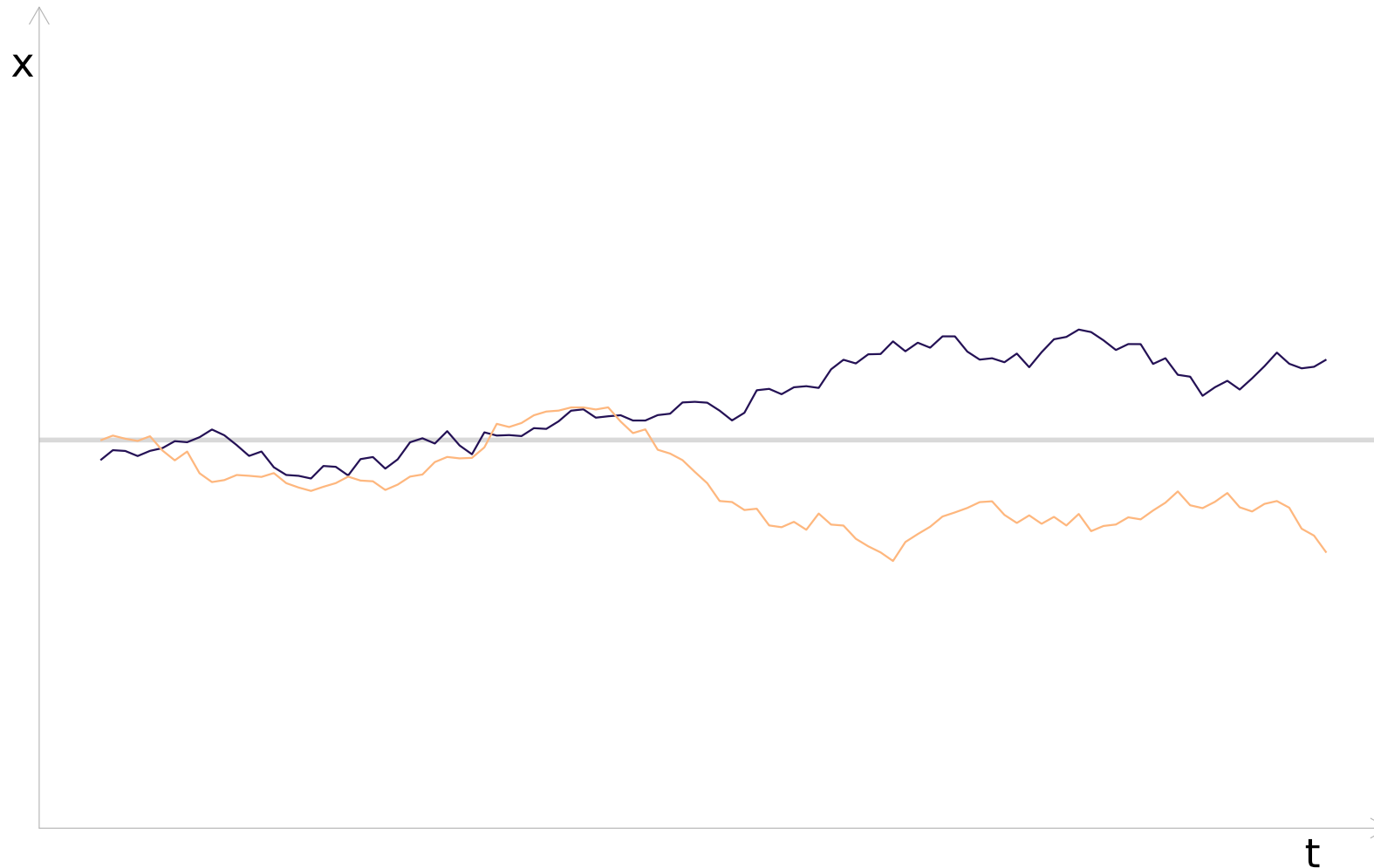
Granger and Newbold simulation example

t statistic ≈ -10.58



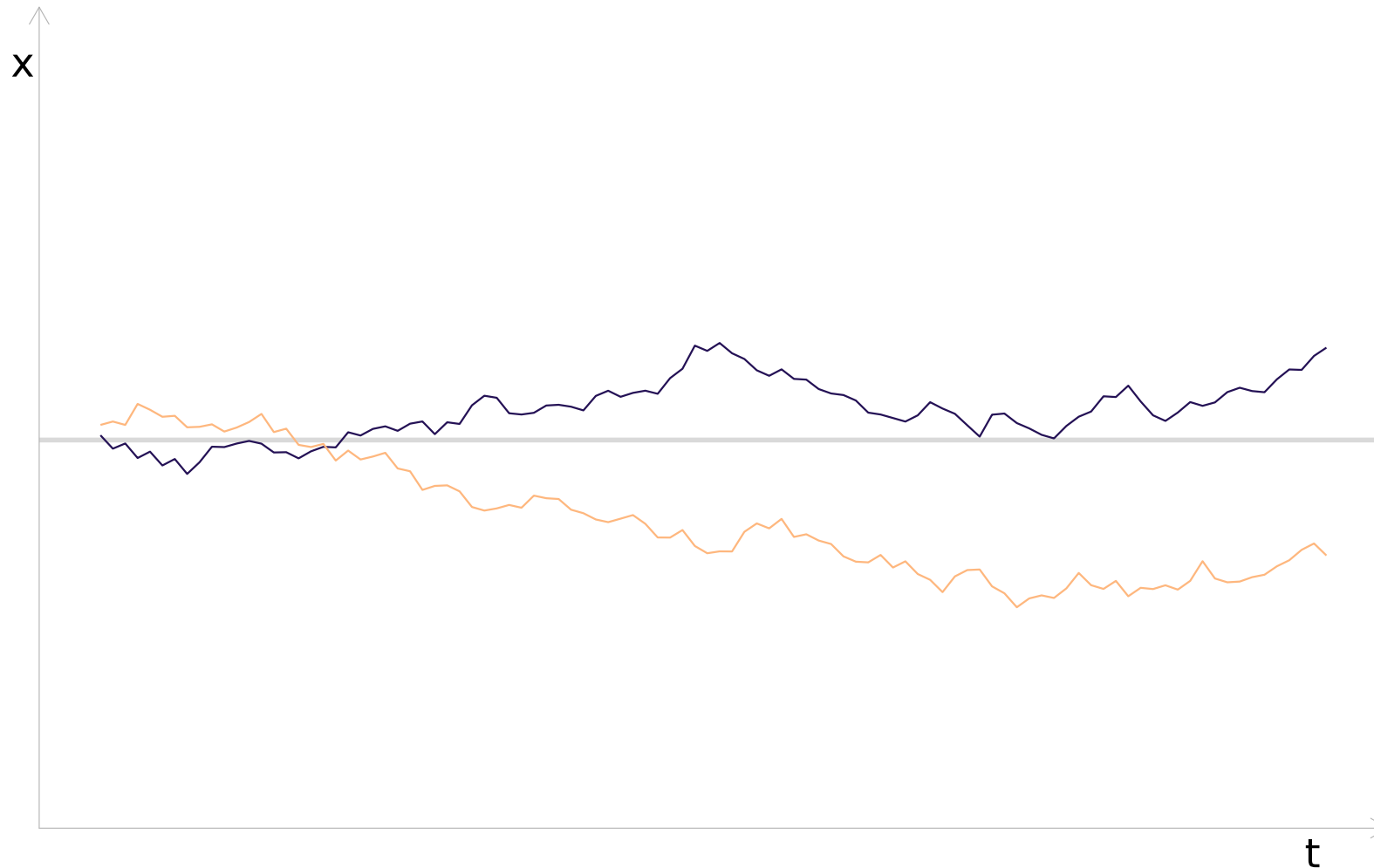
Granger and Newbold simulation example

t statistic ≈ -8.92



Granger and Newbold simulation example

t statistic ≈ -7.23



Nonstationarity

Problem

In our data, 74.6 percent of (independently generated) pairs reject the null hypothesis at the 5% level.

Nonstationarity

A potential solution

Some processes are **difference stationary**, which means we can get back to our stationarity (good behavior) requirement by taking the difference between u_t and u_{t-1} .

Nonstationarity

Testing

Dickey-Fuller and augmented Dickey-Fuller tests are popular ways to test for random walks and other forms of nonstationarity.

1. † People often just test $\beta_1 < 1$.

Summary and conclusion

Stationary time series behave predictably: stable mean, variance, and covariance.

- **Nonstationary processes**, like random walks, can produce **spurious regression results**.
- Differencing can help when the process is **difference stationary**.
- **Bottom line:** test/think carefully before trusting OLS with time series.

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